

KRYVYI RIH, Ukraine

WMO: 337910

Lat: 48.0536N

Lon: 33.2111E

Elev: 124

StdP: 99.84

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.9	-14.9	-20.9	0.6	-17.6	-17.8	0.8	-14.4	11.9	-4.7	10.9	-2.4	3.8	0	0.580

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	33.9	20.3	32.0	19.8	30.1	19.4	22.1	29.8	21.1	28.6	20.3	27.7	4.1	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	14.7	25.2	18.8	13.8	24.1	17.9	13.1	23.2	65.1	30.0	61.9	28.9	59.0	27.6	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.3	9.0	7.9	DB	-21.3	36.1	3.7	1.9	-24.0	37.4	-26.1	38.5	-28.2	39.5	-30.9	40.9	(4)
(5)			WB	-21.6	24.1	3.6	1.5	-24.2	25.2	-26.3	26.0	-28.3	26.8	-30.9	27.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-3.7	-2.4	2.8	10.4	16.4	20.2	22.5	22.0	16.4	9.5	3.5	-1.4	(6)	
	DBStd	10.39	5.71	5.67	4.59	4.02	4.11	3.46	3.12	3.62	4.11	4.61	4.83	5.42	(7)	
	HDD10.0	1667	426	347	224	42	3	0	0	0	3	66	202	354	(8)	
	HDD18.3	3519	684	580	481	238	87	20	3	8	83	275	446	613	(9)	
	CDD10.0	1577	0	0	2	55	201	307	388	373	195	51	6	0	(10)	
	CDD18.3	387	0	0	0	1	27	77	133	122	25	2	0	0	(11)	
	CDH23.3	4006	0	0	0	18	301	753	1349	1310	263	12	0	0	(12)	
	CDH26.7	1521	0	0	0	3	81	254	542	564	76	1	0	0	(13)	
(14) Wind	WSAvg	4.0	4.4	4.4	4.6	4.3	3.8	3.6	3.5	3.6	3.6	3.7	4.1	4.3	(14)	
(15) Precipitation	PrecAvg	494	38	33	36	32	50	61	61	34	42	39	38	36	(15)	
	PrecMax	801	83	91	64	91	86	95	115	85	122	111	82	82	(16)	
	PrecMin	269	8	7	11	0	12	20	16	11	0	1	3	6	(17)	
	PrecStd	109	21	23	17	25	22	23	29	18	32	29	25	24	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.2	10.9	17.9	25.6	31.8	33.2	35.7	36.9	31.9	24.9	17.1	11.2	(19)	
		MCWB	6.9	7.9	10.0	14.6	18.7	20.2	20.9	20.3	18.5	16.0	13.5	9.6	(20)	
	2%	DB	6.1	7.9	14.4	22.2	28.9	31.2	33.6	34.2	28.9	21.6	14.0	9.0	(21)	
		MCWB	5.5	5.8	8.6	12.9	17.5	19.8	20.7	20.1	17.2	14.8	11.8	8.2	(22)	
	5%	DB	4.6	5.9	12.0	20.1	26.8	29.8	31.8	32.1	26.2	18.7	11.9	7.1	(23)	
		MCWB	3.9	4.6	7.1	12.0	16.9	19.5	20.2	19.7	16.2	13.9	10.4	6.4	(24)	
	10%	DB	2.9	4.1	9.6	18.0	24.4	27.9	29.9	30.0	24.0	16.2	10.1	5.4	(25)	
		MCWB	2.3	3.0	6.0	11.0	16.1	18.7	19.9	19.0	15.3	12.6	8.8	4.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.2	8.6	11.0	15.8	20.0	22.7	23.4	23.1	20.2	17.9	14.1	10.1	(27)	
		MCDB	7.7	10.2	16.5	23.2	27.9	29.8	30.6	32.0	27.8	22.7	16.6	10.8	(28)	
	2%	WB	5.5	6.3	9.2	14.0	18.7	21.5	22.2	21.6	18.6	16.0	12.1	8.2	(29)	
		MCDB	6.0	7.4	13.6	20.7	26.2	28.3	29.8	30.0	25.5	19.0	13.9	8.9	(30)	
	5%	WB	3.9	4.6	7.7	12.8	17.7	20.4	21.3	20.7	17.4	14.5	10.7	6.4	(31)	
		MCDB	4.3	5.6	10.8	18.4	24.6	27.1	28.8	28.7	23.4	17.9	11.8	6.9	(32)	
	10%	WB	2.2	3.1	6.3	11.7	16.7	19.5	20.5	19.7	16.3	13.1	9.0	4.9	(33)	
		MCDB	2.6	3.9	8.9	16.9	23.2	25.9	27.8	27.5	22.0	16.1	9.9	5.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	6.1	8.2	10.9	11.6	11.4	11.8	12.0	10.9	9.1	5.9	5.1	(35)	
		MCDBR	5.4	7.8	13.0	14.1	14.3	13.8	14.0	14.5	14.4	12.0	8.5	6.3	(36)	
	5% WB	MCWBR	4.8	6.0	7.8	7.0	6.3	5.7	5.1	5.4	6.4	6.7	6.2	5.4	(37)	
		MCWBR	5.2	7.2	11.2	12.4	12.6	12.1	12.3	12.8	12.1	10.5	7.6	5.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.327	0.357	0.386	0.445	0.416	0.411	0.440	0.434	0.407	0.370	0.353	0.330	(40)	
		taud	2.254	2.206	2.230	2.100	2.270	2.302	2.228	2.236	2.282	2.368	2.329	2.315	(41)	
	Ebn at Noon		720	779	821	804	848	853	821	807	791	763	687	669	(42)	
		Edh at Noon	78	102	117	147	129	126	134	127	111	87	73	66	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.97	1.82	3.04	4.59	5.85	6.25	6.20	5.59	3.87	2.35	1.02	0.70	(44)	
	RadStd		0.19	0.32	0.44	0.42	0.49	0.46	0.34	0.32	0.49	0.32	0.17	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.71	N/A	N/A	+1.24	N/A	N/A	N/A	-156	+140	+100
(47) Regional (10 neighbors)		+0.75	N/A	N/A	+1.10	N/A	N/A	-137	-185	+120	+80

Nomenclature: See separate page